

Question-2.8.1

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Question

If $(-4, 3)$ and $(4, 3)$ are two vertices of an equilateral triangle. Find the coordinates of the third vertex, given that the origin lies in the interior of the triangle.

Solution

Let

$$\mathbf{A} = \begin{pmatrix} -4 \\ 3 \end{pmatrix}, \mathbf{B} = \begin{pmatrix} 4 \\ 3 \end{pmatrix} \text{ and } \mathbf{C} = \begin{pmatrix} x \\ y \end{pmatrix} \quad (1)$$

be the vertices of equilateral triangle.

The direction vector of the segment joining A and B is given by:

$$\mathbf{B} - \mathbf{A} = \begin{pmatrix} 4 - (-4) \\ 3 - 3 \end{pmatrix} = \begin{pmatrix} 8 \\ 0 \end{pmatrix} \quad (2)$$

The side length of equilateral triangle is

$$\|\mathbf{B} - \mathbf{A}\| = \sqrt{(\mathbf{B} - \mathbf{A})^T (\mathbf{B} - \mathbf{A})} \quad (3)$$

$$= \sqrt{(8 \ 0) \begin{pmatrix} 8 \\ 0 \end{pmatrix}} \quad (4)$$

$$= \sqrt{8(8) + 0(0)} \quad (5)$$

$$= 8 \quad (6)$$

Solution

Let **M** be the midpoint of **A** and **B**

$$\mathbf{M} = \frac{\mathbf{A} + \mathbf{B}}{2} \quad (7)$$

$$= \frac{\begin{pmatrix} -4 \\ 3 \end{pmatrix} + \begin{pmatrix} 4 \\ 3 \end{pmatrix}}{2} \quad (8)$$

$$= \frac{\begin{pmatrix} 0 \\ 6 \end{pmatrix}}{2} \quad (9)$$

$$= \begin{pmatrix} 0 \\ 3 \end{pmatrix} \quad (10)$$

Height of equilateral triangle

$$h = \frac{\sqrt{3}}{2} \cdot (8) = 4\sqrt{3} \quad (11)$$

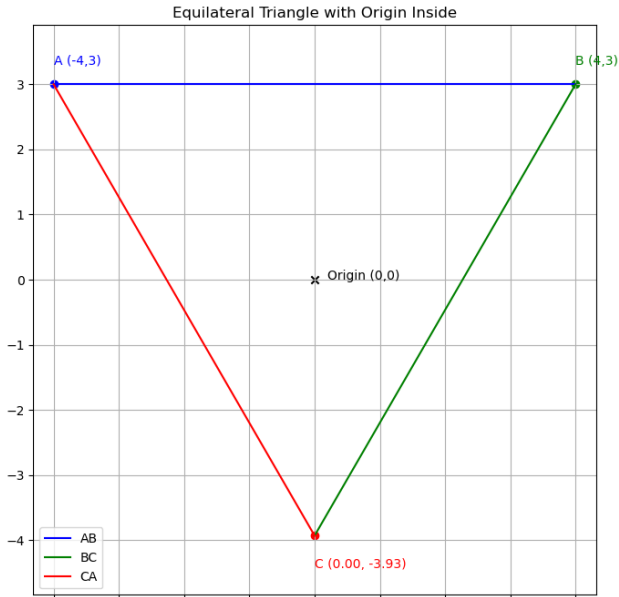
Solution

The perpendicular line drawn from **C** meets the line joining the points **A** and **B** at the midpoint **M**. Thus, the possible values of **C** are

$$\mathbf{C} = \begin{pmatrix} 0 \\ 3 + 4\sqrt{3} \end{pmatrix} \text{ (or) } \begin{pmatrix} 0 \\ 3 - 4\sqrt{3} \end{pmatrix} \quad (12)$$

Since the origin is inside the triangle, choose the lower vertex:

$$\mathbf{C} = \begin{pmatrix} 0 \\ 3 - 4\sqrt{3} \end{pmatrix} \quad (13)$$



```
// code.c
#include <stdio.h>
#include <math.h>

void find_third_vertex(double *x, double *y) {
    // Given two vertices
    double Ax = -4, Ay = 3;
    double Bx = 4, By = 3;

    // Side length AB
    double AB = sqrt((Bx - Ax)*(Bx - Ax) + (By - Ay)*(By - Ay));

    // Midpoint M
    double Mx = (Ax + Bx) / 2.0;
    double My = (Ay + By) / 2.0;

    // Height for equilateral triangle
    double h = (sqrt(3) / 2.0) * AB;
```

```
// Two possible third vertices (perpendicular from midpoint)
double C1x = Mx;
double C1y = My + h;

double C2x = Mx;
double C2y = My - h;

// Choose the third vertex such that origin lies inside
triangle
// Since origin is inside, pick vertex with y < 0
if (C1y < 0) {
    *x = C1x;
    *y = C1y;
} else {
    *x = C2x;
    *y = C2y;
}
}
```



```
# Code by GVV Sharma  
# Modified for Problem Solution  
# Released under GNU GPL  
# Calculating area enclosed between curves  
# call.py  
from ctypes import CDLL, c_double, POINTER  
  
# Load the shared library  
lib = CDLL('./libtriangle.so')  
  
# Setup argument types  
lib.find_third_vertex.argtypes = [POINTER(c_double), POINTER(  
    c_double)]
```

```
def find_third_vertex():
    x = c_double()
    y = c_double()
    lib.find_third_vertex(x, y)
    return x.value, y.value

if __name__ == "__main__":
    cx, cy = find_third_vertex()
    print(f"Third vertex coordinates from C code: ({cx}, {cy})")
```

```
import numpy as np
import matplotlib.pyplot as plt
from ctypes import CDLL, c_double, POINTER

# Load the shared library
lib = CDLL('./libtriangle.so')

# Setup argument types for C function
lib.find_third_vertex.argtypes = [POINTER(c_double), POINTER(
    c_double)]

def get_third_vertex():
    x = c_double()
    y = c_double()
    lib.find_third_vertex(x, y)
    return x.value, y.value
```

```
# Given vertices
A = np.array([-4, 3])
B = np.array([4, 3])
C = np.array(get_third_vertex())
Origin = np.array([0, 0])

# Plot the triangle edges
plt.figure(figsize=(7, 7))
plt.plot([A[0], B[0]], [A[1], B[1]], 'b-', label='AB')
plt.plot([B[0], C[0]], [B[1], C[1]], 'g-', label='BC')
plt.plot([C[0], A[0]], [C[1], A[1]], 'r-', label='CA')

# Mark and label points
plt.scatter(*A, color='blue')
plt.text(A[0], A[1]+0.3, "A (-4,3)", color='blue')

plt.scatter(*B, color='green')
plt.text(B[0], B[1]+0.3, "B (4,3)", color='green')
```

```
plt.scatter(*C, color='red')
plt.text(C[0], C[1]-0.5, f"C ({C[0]:.2f}, {C[1]:.2f})", color='red')
plt.scatter(*Origin, color='black', marker='x')
plt.text(Origin[0]+0.2, Origin[1], 'Origin (0,0)', color='black')

plt.title("Equilateral Triangle with Origin Inside")
plt.legend()
plt.grid(True)
plt.axis('equal')
plt.tight_layout()
plt.show()
```